

# Smart Museum Artifact Conservation System

## Background

A museum has a special chamber used to protect extremely valuable historical artifacts. The chamber contains sensors and an automated environmental-control system.

The system continuously monitors:

- temperature
- humidity
- light exposure
- vibration
- chamber door status
- artifact condition
- power availability

The objective is not simply to keep the room comfortable. The system must maintain **specific environmental conditions for the artifact** and react appropriately when abnormal conditions occur.

## Scenario

When the conservation chamber is powered on, the system performs a self-check of its sensors and environmental-control devices. The chamber cannot enter normal conservation mode until all essential sensors are working correctly.

Once the self-check succeeds, the chamber enters **MONITORING** mode.

When an artifact is placed inside the chamber, the system records the artifact's identification information and its required environmental limits. The system then begins monitoring the artifact.

The chamber door must be closed before active conservation can begin. If the door remains open, the system continues monitoring but does not activate the normal conservation process.

When the door is closed and the artifact's required environmental profile has been loaded, the system enters **CONSERVATION\_ACTIVE** mode.

During conservation, the system continuously compares the actual temperature and humidity with the limits specified for the artifact.

If the temperature moves outside the permitted range, the system attempts to restore the required temperature using its environmental-control mechanism.

If the humidity moves outside the permitted range, the system similarly attempts to correct the humidity.

However, the system must not immediately declare the artifact safe after issuing a correction command. It must verify through sensor readings that the environmental condition has actually returned to the permitted range.

If an environmental condition cannot be corrected within the allowed recovery period, the system enters **PROTECTION\_MODE**. In this mode, the system prioritizes protecting the artifact rather than maintaining normal conservation operation.

While in protection mode, the system may reduce light exposure and activate additional environmental controls. It also generates an alert for the museum operator.

The chamber contains a vibration sensor. If significant vibration is detected while an artifact is inside, the system must temporarily suspend activities that could increase the risk to the artifact and enter **VIBRATION\_RESPONSE**.

The system cannot return to normal conservation merely because the vibration has stopped. It must first verify that the vibration level has remained below the permitted threshold for the required stabilization period.

If the chamber door is opened while conservation is active, the system must immediately suspend normal conservation activities. It must not continue normal environmental operation while the chamber is open.

When the door is closed again, the system does not automatically resume conservation. It must verify the artifact's environmental conditions and sensor status before returning to normal operation.

If the system loses power during conservation, it switches to an emergency power source if available. If emergency power is unavailable, it records the incident and enters a safe shutdown state.

An operator may remove the artifact only when the system confirms that the chamber is in a safe condition and no active protection response is underway.

### **Task 1: Identify Operations**

Read the scenario carefully and identify at least 12 operations performed by the system. Do not use state names as operation names.

### **Student Task 2**

Create Complete Operation Schema

